

# CMPU2032 - Programming & Algorithms 2 (2025/26)

## *Project Report*

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## Overview

### Assignment Description

A hospital wants to digitize its patient handling system. You are hired to develop a Python-based system that efficiently manages patient records, emergency cases, and data processing using appropriate data structures and algorithms.

### Problem Statement

Design and implement a menu-driven Python program that demonstrates the use of following requirements:

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## Explanation of the data structures used

There were three core structures used to manage the "Smart Hospital System" data. The breakdown of these are listed below: -

### 1. Patient Data-Class

- Simple data structure `Patient`, used to store patient details loaded into the system.
- Keep the details to the basics: `id`, `name`, `age` & `condition`.
- Included data-type validation to ensure `id` and `age` are integers.

### 2. Patient Records Linked-List

- The `PatientRecords` class implements a simple linked-list. This class keeps track of the `head` and `tail` node in the linked list.
- Uses a `head` node to track the start of the list, with `tail` node tracking the end of the list.
- There is a `LinkNode` data-class that is used to manage/store each node in the linked-list.
- The `LinkNode` will hold the `Patient` record as-well-as the next `LinkNode` in the list (if applicable).
- This linked-list can be accessed via the following operations: -
  - *Add Patient* `add_patient(patient: Patient) -> bool`: adds a new `Patient` record to the end of the linked list. Returns a `boolean` to the calling program to indicate if this was completed successfully.
  - *Delete Patient* `delete_patient(patient_id: int) -> Patient | None`: removes a `Patient` record from the list based on the provided `patient_id`. This will also ensure that the list linking is maintained. The removed `Patient` record (or `None` if not found) is returned to the calling program.
  - *Find Patient* `find_patient(patient_id: int) -> Patient | None`: using the provided `ID`, find and return the respective `Patient` record. This will return the found `Patient` record or `None` if not found.
  - *List Patients* `list_patients() -> list[Patient]`: returns all the `Patient` records in the linked-list as an `Array`.
  - *Clear List* `clear_list()`: clears all the `Patient` records and resets the linked-list.
  - *Next ID* `next_id() -> int`: generates a random 4-digit ID for each new `Patient` record.

### 3. Emergency Queue

- The `EmergencyQueue` class implements a simple queue. Uses an array to queue each `Patient` record added to the queue. New records will be added to the end of the array (enqueue), whilst the next `Patient` served will be taken from the start of the array. Thus, implementing a First-in, First-out (FIFO) queue.
- This queue can be accessed via the following operations: -
  - *Enqueue Patient* `enqueue_patient(patient: Patient) -> bool`: adds (enqueue) a `Patient` record to the end of the queue. There is a check to ensure that the `Patient` is not already in the queue. Will return a `boolean` to the calling program if the operation is completed successfully.
  - *Dequeue Patient* `dequeue_patient() -> Patient | None`: removes (dequeue) the first `Patient` record from the queue. Returns the respective `Patient` record (or `None` if queue is empty) to the calling program.
  - *View Queue* `view_queue() -> list[Patient]`: returns all the `Patient` records in the linked-list as an `Array`.
  - *Is Empty* `is_empty()`: checks if the queue state is currently *empty* and will return a `boolean` result to the respective calling program.

### Description of sorting algorithms implemented

Per the requirements for this project, two sorting algorithms needed to be implemented and used with different use cases. These implementations will respectively sort the `PatientRecords` linked-list. Both implementations support dynamic sorting by any patient attribute through a configurable `key` parameter.

**Merge Sort:** uses a divide-and-conquer approach ...

- Will recursively divide the list into halves (used a slow/fast pointer technique to find the middle of the linked-list)
- Then sort each half independently before merging the halves back together while maintaining order
- Time Complexity:  $O(\log n)$

**Shell Sort:** uses gap-based insertion sort approach ...

- Needed to first convert the linked-list to an array to be able to perform this sort
- Will perform multiple passes with decreasing gap intervals
- Gap reduces by half each iteration until gap equals 1
- Time Complexity:  $O(\log n)$  to  $O(n^2)$

## Test cases

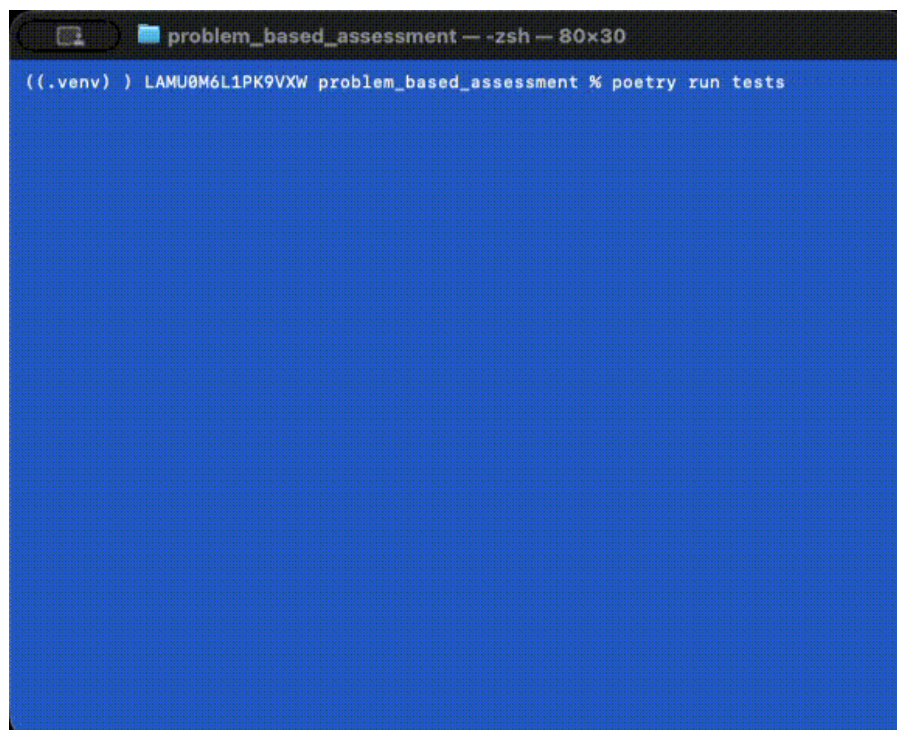
I have added several tests to the `tests` folder in this project. These tests have been created to validate several core functions of the "Smart Health System" application. Have coded test cases that cover positive and negative use-case scenarios.

Each test case is an insular unit-test (only testing the functionality of the respective target class). When there are any dependencies on other objects (for example, adding a patient depends on the patient records linked-list) a MOCK is used to simulate the external operations.

Some examples of the tests created are: -

- `test_patient.py`: covers test cases to verify normal `Patient` creation and "to string" conversion, as-well-as negative scenario where an invalid age is entered.
- `test_emergency_queue.py`: covers test cases for several of the `EmergencyQueue` class operations (would have scenarios for every function in a real application). Additionally, have test cases to verify negative scenarios, like adding a `Patient` to the queue twice.
- `test_add_patient.py`: similar coverage of `AddPatient` class operations is provided by in these test cases, with negative test to ensure duplicate `Patient` records cannot get added to the linked-list.

The recording below shows how the tests can be executed.



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## Explanation of the design pattern selected

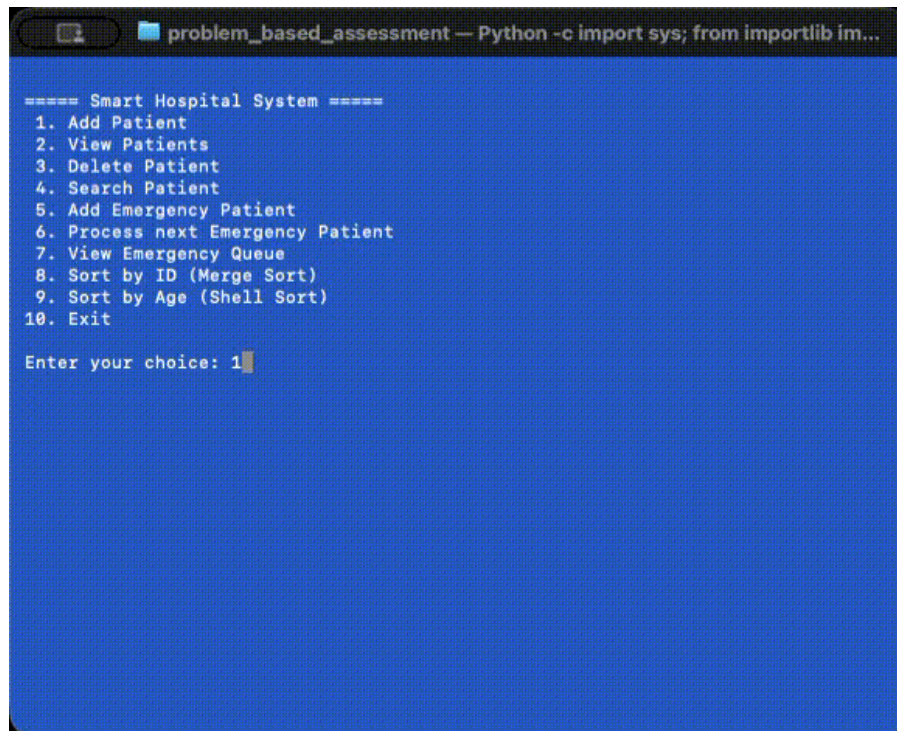
I couldn't find any good design-pattern use-cases for this project as its simplicity doesn't require such additional complexity.

One area I thought could use a **Singleton** design-pattern, was for the emergency queue (**EmergencyQueue**) and patient-records linked-list (**PatientRecords**). This would simplify the need to pass an instance of these class objects to several other (e.g. main-menu & menu option) classes but would make unit-testing more complex.

I did use the **Builder** (a creational design-pattern) to construct **MainMenu** class object. This pattern is implemented in **MenuBuilder** class and breaks down the creation of an object into separate steps. This would typically be more suitable to use-cases where there would be multiple instances of the class object with varying configurations. This reduces the need to have multiple constructors for different uses or a very complex constructor that takes a large number of attributes.

## Sample outputs of the program

- Adding a new patient to the linked-list



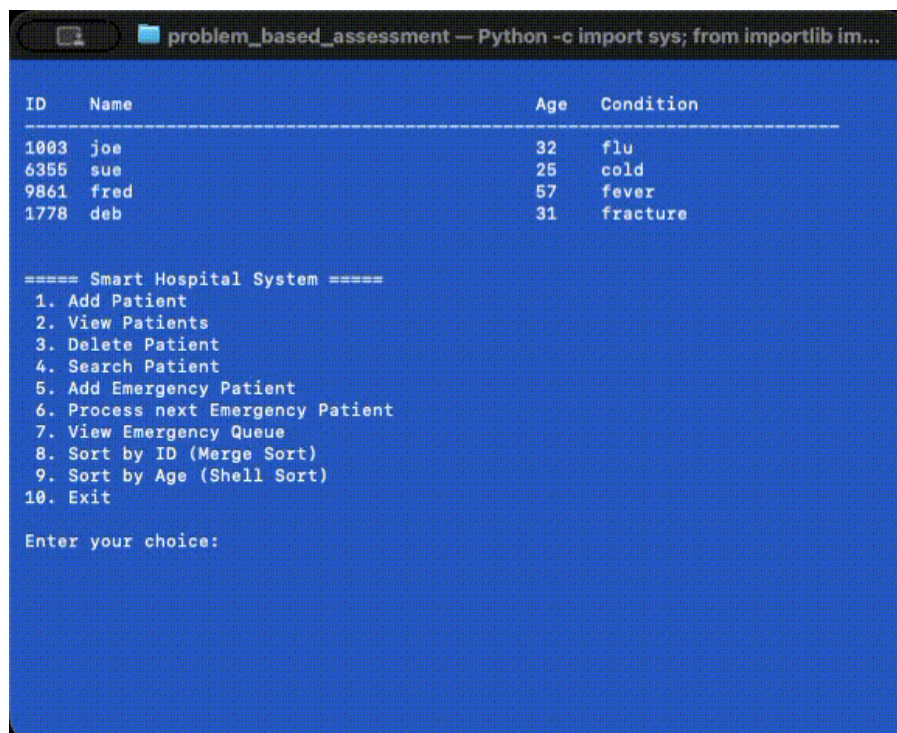
```
problem_based_assessment — Python -c import sys; from importlib im...

===== Smart Hospital System =====
1. Add Patient
2. View Patients
3. Delete Patient
4. Search Patient
5. Add Emergency Patient
6. Process next Emergency Patient
7. View Emergency Queue
8. Sort by ID (Merge Sort)
9. Sort by Age (Shell Sort)
10. Exit

Enter your choice: 1
```

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- Removing an existing patient from the linked-list



| ID   | Name | Age | Condition |
|------|------|-----|-----------|
| 1003 | joe  | 32  | flu       |
| 6355 | sue  | 25  | cold      |
| 9861 | fred | 57  | fever     |
| 1778 | deb  | 31  | fracture  |

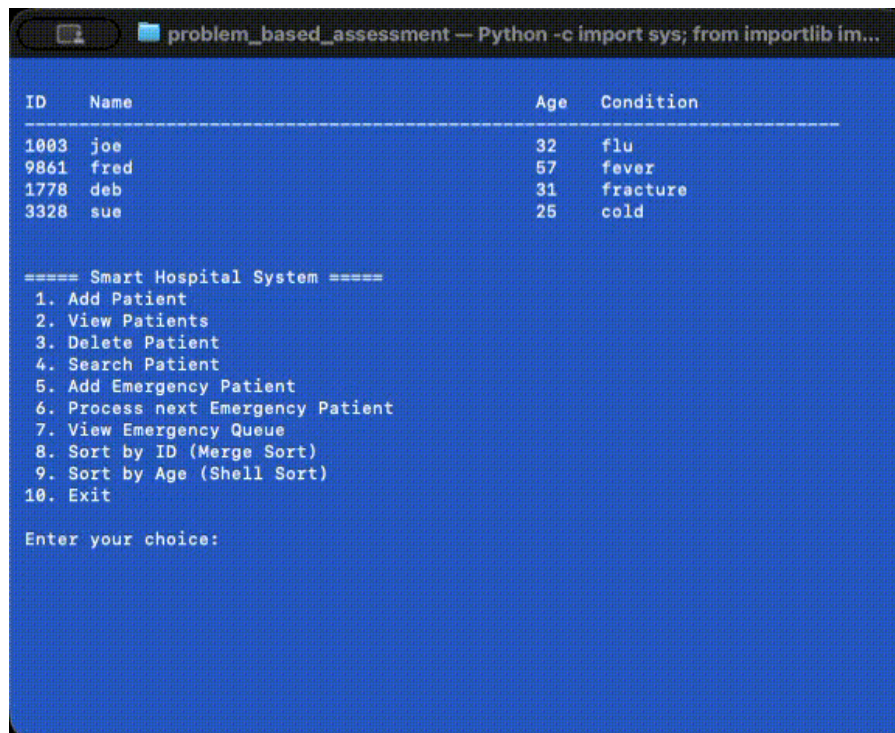
```
===== Smart Hospital System =====
1. Add Patient
2. View Patients
3. Delete Patient
4. Search Patient
5. Add Emergency Patient
6. Process next Emergency Patient
7. View Emergency Queue
8. Sort by ID (Merge Sort)
9. Sort by Age (Shell Sort)
10. Exit

Enter your choice:
```

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- Adding a patient to the emergency-queue



```
problem_based_assessment — Python -c import sys; from importlib im...

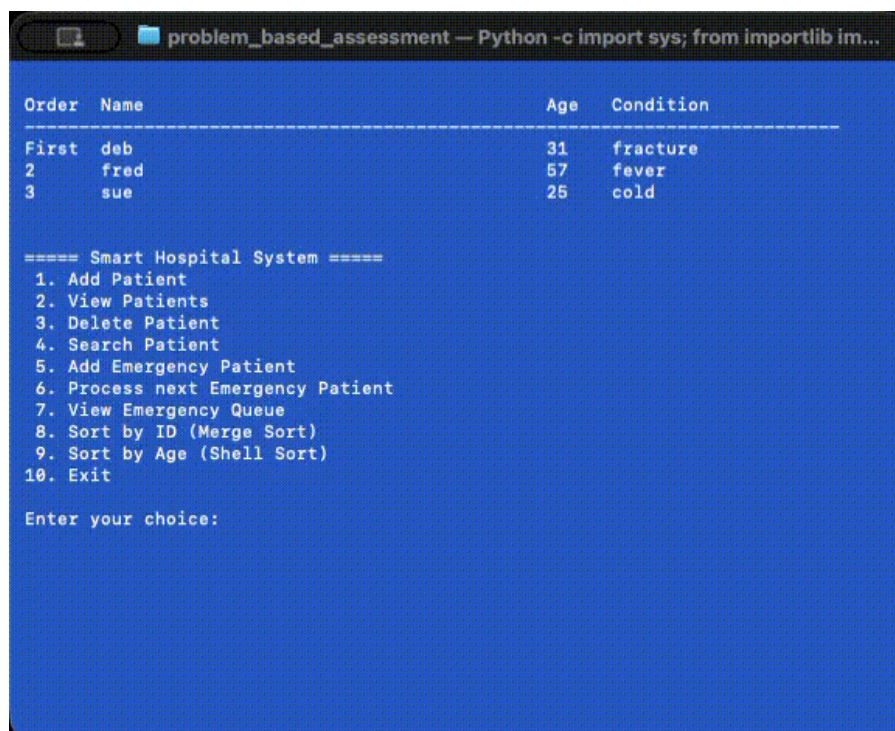
ID      Name      Age      Condition
-----
1003    joe      32      flu
9861    fred     57      fever
1778    deb      31      fracture
3328    sue      25      cold

===== Smart Hospital System =====
1. Add Patient
2. View Patients
3. Delete Patient
4. Search Patient
5. Add Emergency Patient
6. Process next Emergency Patient
7. View Emergency Queue
8. Sort by ID (Merge Sort)
9. Sort by Age (Shell Sort)
10. Exit

Enter your choice:
```

Animated GIF cannot be viewed in generate PDF file, alternatively download/view the recording [here](#).

- Serving the next patient in the emergency-queue



```
problem_based_assessment — Python -c import sys; from importlib im...

Order   Name      Age      Condition
-----
First   deb      31      fracture
2       fred     57      fever
3       sue      25      cold

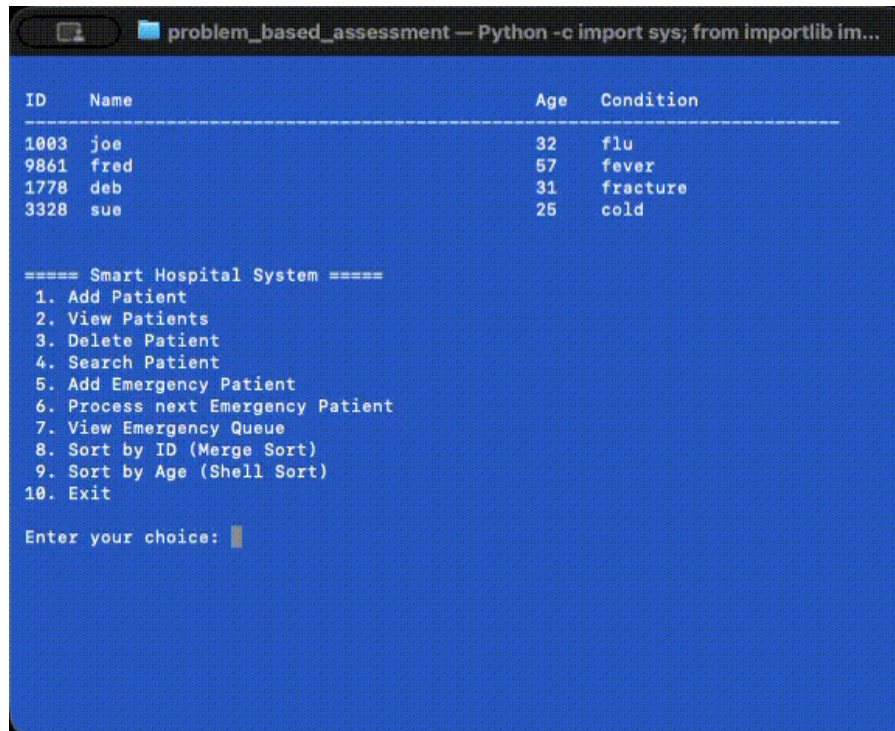
===== Smart Hospital System =====
1. Add Patient
2. View Patients
3. Delete Patient
4. Search Patient
5. Add Emergency Patient
6. Process next Emergency Patient
7. View Emergency Queue
8. Sort by ID (Merge Sort)
9. Sort by Age (Shell Sort)
10. Exit

Enter your choice:
```

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- Sorting the linked-list by ID



```
problem_based_assessment — Python -c import sys; from importlib im...

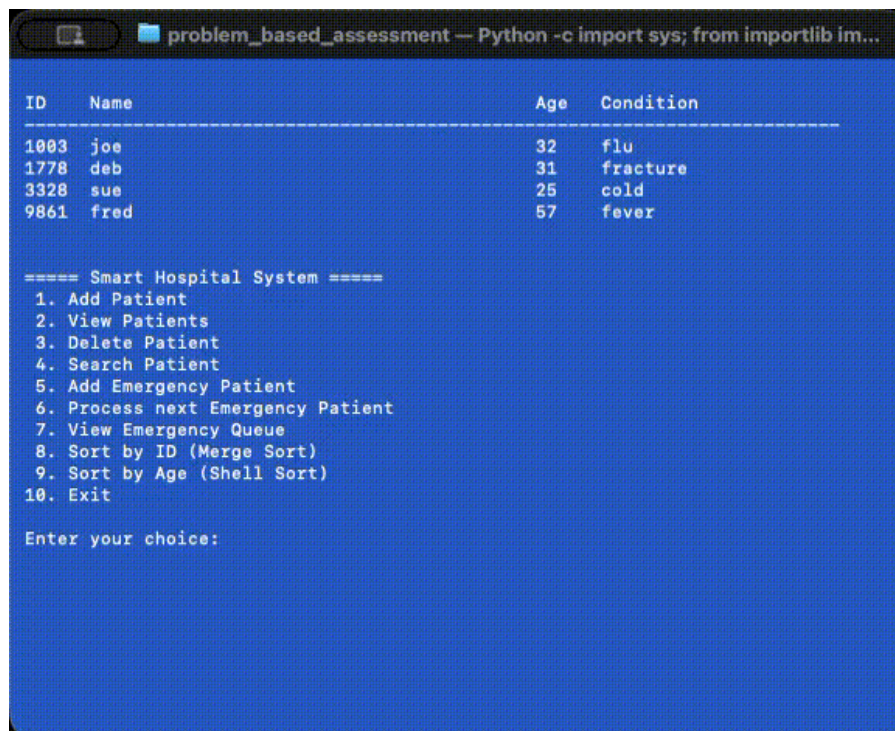
ID      Name      Age      Condition
-----
1003    joe      32      flu
9861    fred     57      fever
1778    deb      31      fracture
3328    sue      25      cold

===== Smart Hospital System =====
1. Add Patient
2. View Patients
3. Delete Patient
4. Search Patient
5. Add Emergency Patient
6. Process next Emergency Patient
7. View Emergency Queue
8. Sort by ID (Merge Sort)
9. Sort by Age (Shell Sort)
10. Exit

Enter your choice: 
```

Animated GIF cannot be viewed in generate PDF file, alternatively download/view the recording [here](#).

- Sorting the linked-list by AGE



```
problem_based_assessment — Python -c import sys; from importlib im...

ID      Name      Age      Condition
-----
1003    joe      32      flu
1778    deb      31      fracture
3328    sue      25      cold
9861    fred     57      fever

===== Smart Hospital System =====
1. Add Patient
2. View Patients
3. Delete Patient
4. Search Patient
5. Add Emergency Patient
6. Process next Emergency Patient
7. View Emergency Queue
8. Sort by ID (Merge Sort)
9. Sort by Age (Shell Sort)
10. Exit

Enter your choice: 
```

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## Setup and Execution

### Create Virtual Environment

To ensure this project does not impact any other parts of your system, a virtual environment should be setup. The assumption is that you have Python (version 3.11 or later) installed on your computer. This can simple be done by executing the following commands: -

- Windows ...

```
python -m venv .venv
.venv\Scripts\activate
```

- Mac, Linux or WSL ...

```
python3 -m venv .venv
source .venv/bin/activate
```

This can be done in other ways, for example: through your IDE or with tools like [pipx](#). Please refer to the respective documentation for any other approach you wish to follow.

### Install dependencies

This project uses [Poetry](#) for dependency management, building, running and testing. Run the commands below to install [Poetry](#) into your Python virtual-environment as-well-as all the required project dependencies: -

```
pip3 install poetry
poetry install
```

---

## Launching the Smart Hospital System Application

To run the command-line interface: -

```
poetry run main
```

## Executing all Application Code Unit-Tests

To run the unit tests: -

```
poetry run tests
```